Applications and Limitations of Strategic Translation in Selection Principles

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Context

We will be considering the usual selection principles $\mathsf{S}_1(\mathcal{A}, \mathcal{B})$ and $\mathsf{S}_{\mathrm{fin}}(\mathcal{A}, \mathcal{B})$; we also consider the corresponding two-player games $\mathsf{G}_1(\mathcal{A}, \mathcal{B})$ and $\mathsf{G}_{\mathrm{fin}}(\mathcal{A}, \mathcal{B})$ of length ω. The players will be named One and Two.

Motivation

- Selection Principles themselves are equivalent to the negation of a particular kind of limited-information strategy. Some cornerstone results are the equivalence of perfect-information strategies and this kind of limited-information strategy for the first player (theorems of Hurewicz, Pawlikowsi, and Scheepers).
- The study of limited-information strategies and game duality helped answer some questions related to the property of closed discrete selection in the C_p context. In particular, Clontz and Holshouser characterized when the second player, the point-picker, has a winning strategy in the closed discrete selection game on $C_p(X)$.

Motivation

- Many arguments in the literature for certain implications between perfect-information strategies and selection principles have the same structure, where the only difference is the kind of strategy that is being considered.

Inspiration in $C_p(X)$

A common theme when establishing relationships between, for example, $\mathsf{S}_1(\mathcal{A}, \mathcal{B})$ and $\mathsf{S}_1(\mathcal{A}', \mathcal{B}')$, is to draw out structural relationships between the underlying topological constructions. For example, when establishing relationships between properties of a Tychonoff space X and $C_p(X)$, we can look at some "ground level" correspondences:

$$f \mapsto W_n(f) := f^{\leftarrow}[(-2^{-n}, 2^{-n})], C_p(X) \to \mathscr{T}_X,$$

and

$$U \mapsto \mathscr{D}^*(U) := \{f \in C_p(X) : f \restriction_{X \setminus U} \equiv \mathbf{1}\}, \mathscr{T}_X \to \wp(C_p(X)).$$

These correspondences are used to connect the ω-Rothberger game, $\mathsf{G}_1(\Omega_X, \Omega_X)$, on X to clustering games on $C_p(X)$, which are related to the Gruenhage W-games and the closed discrete selection game of (Guerrero Sánchez and) Tkachuk.

- In particular, if $\mathscr{F} \subseteq C_p(X)$ is such that $\mathbf{0} \in \mathrm{cl}_{C_p(X)} \mathscr{F}$, then $W_n[\mathscr{F}] \in \Omega_X$.
- On the other hand, if $\mathscr{U} \in \Omega_X$, then $\bigcup \mathscr{D}^*[\mathscr{U}]$ is dense in $C_p(X)$.

Strategy Types

- Constant (winning) strategies for One. The non-existence of such a strategy is equivalent to Lindelöf-like properties.
- Pre-determined (winning) strategies for One. The non-existence of such a strategy is equivalent to the corresponding selection principle.

  (We use the term "pre-determined" since One is playing a fixed sequence decided before gameplay, and not paying any attention to Two's moves at all.)
- (Full/Perfect-information) Winning strategies for One.
- (Full/Perfect-information) Winning strategies for Two.
- Markov (winning) strategies for Two. These are typically seen as "traditional" topological properties.

Notation

If $\mathcal{G}$ is a selection game,

- $\mathrm{I} \underset{\text{cnst}}{\uparrow} \mathcal{G}$ means One has a constant winning strategy in $\mathcal{G}$.
- $\mathrm{I} \underset{\text{pre}}{\uparrow} \mathcal{G}$ means One has a pre-determined winning strategy in $\mathcal{G}$.
- $\mathrm{I} \uparrow \mathcal{G}$ means One has a winning strategy in $\mathcal{G}$.
- $\mathrm{II} \uparrow \mathcal{G}$ means Two has a winning strategy in $\mathcal{G}$.
- $\mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{G}$ means Two has a Markov winning strategy in $\mathcal{G}$.

Note

$$\mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{G} \implies \mathrm{II} \uparrow \mathcal{G} \implies \mathrm{I} \not\uparrow \mathcal{G} \implies \mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathcal{G} \implies \mathrm{I} \underset{\text{cnst}}{\not\uparrow} \mathcal{G}$$

A Partial Order

For selection games $\mathcal{G}$ and $\mathcal{H}$, we write $\mathcal{G} \leqslant_{\mathrm{II}} \mathcal{H}$ to mean

$$\begin{cases} \mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{G} & \Longrightarrow & \mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{H} \\ \mathrm{II} \uparrow \mathcal{G} & \Longrightarrow & \mathrm{II} \uparrow \mathcal{H} \\ \mathrm{I} \not\uparrow \mathcal{G} & \Longrightarrow & \mathrm{I} \not\uparrow \mathcal{H} \\ \mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathcal{G} & \Longrightarrow & \mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathcal{H} \end{cases}$$

If, in addition,

$$\mathrm{I} \underset{\text{cnst}}{\not\uparrow} \mathcal{G} \Longrightarrow \mathrm{I} \underset{\text{cnst}}{\not\uparrow} \mathcal{H},$$

then we write $\mathcal{G} \leqslant_{\mathrm{II}}^{+} \mathcal{H}$.

Note that, for $\mathsf{G}_\square(\mathcal{A}', \mathcal{B}')$ and $\mathsf{G}_\square(\mathcal{A}, \mathcal{B})$, the implication

$$\mathrm{I} \underset{\text{pre}}{\not\vdash} \mathsf{G}_\square(\mathcal{A}', \mathcal{B}') \implies \mathrm{I} \underset{\text{pre}}{\not\vdash} \mathsf{G}_\square(\mathcal{A}, \mathcal{B})$$

is equivalent to

$$\mathsf{S}_\square(\mathcal{A}', \mathcal{B}') \implies \mathsf{S}_\square(\mathcal{A}, \mathcal{B}).$$

The Translation Idea

In full generality, translation functions are player- and turn-based. The general idea is as follows:

$\mathsf{G}_{\text{fin}}(\mathcal{A}', \mathcal{B}')$

I II

$\mathsf{G}_{\text{fin}}(\mathcal{A}, \mathcal{B})$

I II

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I	II

$\mathsf{G}_{\text{fin}}(\mathcal{A}, \mathcal{B})$

I	II
$\mathscr{U}_0$	

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$$\begin{array}{ccc} \mathsf{G}_{\mathrm{fin}}(\mathcal{A}', \mathcal{B}') & & \mathsf{G}_{\mathrm{fin}}(\mathcal{A}, \mathcal{B}) \\ \underline{\quad \mathrm{I} \qquad\qquad \mathrm{II} \quad} & & \underline{\quad \mathrm{I} \qquad\qquad \mathrm{II} \quad} \\ \overleftarrow{\mathsf{T}}_{\mathrm{I}}(\mathscr{U}_0, 0) & \longleftarrow & \mathscr{U}_0 \end{array}$$

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I	II
$\overleftarrow{\mathsf{T}}_{\mathrm{I}}(\mathscr{U}_0, 0)$	$\mathscr{F}_0$

$\mathsf{G}_{\text{fin}}(\mathcal{A}, \mathcal{B})$

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$$\begin{array}{ccc}
\mathsf{G}_{\mathrm{fin}}(\mathcal{A}', \mathcal{B}') & & \mathsf{G}_{\mathrm{fin}}(\mathcal{A}, \mathcal{B}) \\
\mathrm{I} \qquad\qquad \mathrm{II} & & \mathrm{I} \qquad\qquad \mathrm{II} \\
\overleftarrow{\mathsf{T}}_{\mathrm{I}}(\mathscr{U}_0, 0) & \mathscr{F}_0 \xrightarrow{\mathscr{U}_0} & \overrightarrow{\mathsf{T}}_{\mathrm{II}}(\mathscr{F}_0, \mathscr{U}_0, 0)
\end{array}$$

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I	II
$\overleftarrow{\mathsf{T}}_{\text{I}}(\mathscr{U}_0, 0)$	
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$\mathsf{G}_{\text{fin}}(\mathcal{A}, \mathcal{B})$

I	II
$\mathscr{U}_0$	
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$\overleftarrow{\mathsf{T}}_{\text{I}}(\mathscr{U}_1, 1)$	

$\mathsf{G}_{\text{fin}}(\mathcal{A}, \mathcal{B})$

I	II
$\mathscr{U}_0$	$\overrightarrow{\mathsf{T}}_{\text{II}}(\mathscr{F}_0, \mathscr{U}_0, 0)$
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I	II
$\mathscr{U}_0$	
	$\overrightarrow{\mathsf{T}}_{\text{II}}(\mathscr{F}_0, \mathscr{U}_0, 0)$
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$\mathsf{G}_{\text{fin}}(\mathcal{A}, \mathcal{B})$

I	II
$\mathscr{U}_0$	
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$\mathscr{U}_1$	
	$\overrightarrow{\mathsf{T}}_{\text{II}}(\mathscr{F}_1, \mathscr{U}_1, 1)$
$\mathscr{U}_2$	

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	$\mathscr{F}_0$
$\overleftarrow{\mathsf{T}}_{\text{I}}(\mathscr{U}_1, 1)$	
	$\mathscr{F}_1$
$\overleftarrow{\mathsf{T}}_{\text{I}}(\mathscr{U}_2, 2)$	

$\mathsf{G}_{\text{fin}}(\mathcal{A}, \mathcal{B})$

I	II
$\mathscr{U}_0$	
	$\overrightarrow{\mathsf{T}}_{\text{II}}(\mathscr{F}_0, \mathscr{U}_0, 0)$
$\mathscr{U}_1$	
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	$\mathscr{F}_1$
$\overleftarrow{\mathsf{T}}_{\text{I}}(\mathscr{U}_2, 2)$	
	$\mathscr{F}_2$

$\mathsf{G}_{\text{fin}}(\mathcal{A}, \mathcal{B})$

I	II
$\mathscr{U}_0$	
	$\overrightarrow{\mathsf{T}}_{\text{II}}(\mathscr{F}_0, \mathscr{U}_0, 0)$
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	$\mathscr{F}_1$
$\overleftarrow{\mathsf{T}}_{\text{I}}(\mathscr{U}_2, 2)$	
	$\mathscr{F}_2$

$\mathsf{G}_{\text{fin}}(\mathcal{A}, \mathcal{B})$

I	II
$\mathscr{U}_0$	
	$\overrightarrow{\mathsf{T}}_{\text{II}}(\mathscr{F}_0, \mathscr{U}_0, 0)$
$\mathscr{U}_1$	
	$\overrightarrow{\mathsf{T}}_{\text{II}}(\mathscr{F}_1, \mathscr{U}_1, 1)$
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	$\mathscr{F}_0$
$\overleftarrow{\mathsf{T}}_{\text{I}}(\mathscr{U}_1, 1)$	
	$\mathscr{F}_1$
$\overleftarrow{\mathsf{T}}_{\text{I}}(\mathscr{U}_2, 2)$	
	$\mathscr{F}_2$
$\vdots$	

$\mathsf{G}_{\text{fin}}(\mathcal{A}, \mathcal{B})$

I	II
$\mathscr{U}_0$	
	$\overrightarrow{\mathsf{T}}_{\text{II}}(\mathscr{F}_0, \mathscr{U}_0, 0)$
$\mathscr{U}_1$	
	$\overrightarrow{\mathsf{T}}_{\text{II}}(\mathscr{F}_1, \mathscr{U}_1, 1)$
$\mathscr{U}_2$	
	$\overrightarrow{\mathsf{T}}_{\text{II}}(\mathscr{F}_2, \mathscr{U}_2, 2)$
	$\vdots$

Formally, a translation from $\mathsf{G}_{\mathrm{fin}}(\mathcal{A},\mathcal{B})$ to $\mathsf{G}_{\mathrm{fin}}(\mathcal{A}',\mathcal{B}')$ is a pair of functions $\overleftarrow{\mathsf{T}}_{\mathrm{I}} : \mathcal{A} \times \omega \to \mathcal{A}'$ and

$$\overrightarrow{\mathsf{T}}_{\mathrm{II}} : \left[\bigcup \mathcal{A}'\right]^{<\omega} \times \mathcal{A} \times \omega \to \left[\bigcup \mathcal{A}\right]^{<\omega}$$

where, for each $\mathscr{U} \in \mathcal{A}$, $\mathscr{F} \in \left[\overleftarrow{\mathsf{T}}_{\mathrm{I}}(\mathscr{U}, n)\right]^{<\omega}$, and $n \in \omega$, $\overrightarrow{\mathsf{T}}_{\mathrm{II}}(\mathscr{F}, \mathscr{U}, n) \in [\mathscr{U}]^{<\omega}$.

The natural modifications can be made for the single-selection games.

A Translation Theorem (C. & Holshouser)

Given $\mathsf{G}_{\text{fin}}(\mathcal{A},\mathcal{B})$ and $\mathsf{G}_{\text{fin}}(\mathcal{A}',\mathcal{B}')$, suppose that we have translation functions $\overleftarrow{\mathsf{T}}_{\text{I}}$ and $\overrightarrow{\mathsf{T}}_{\text{II}}$ such that, for every $\mathscr{U} \in \mathcal{A}^\omega$ and

$$\mathscr{F} \in \prod_{n\in\omega} \left[\overleftarrow{\mathsf{T}}_{\text{I}}(\mathscr{U}_n, n)\right]^{<\omega},$$

$$\bigcup_{n\in\omega} \mathscr{F}_n \in \mathcal{B}' \implies \bigcup_{n\in\omega} \overrightarrow{\mathsf{T}}_{\text{II}}(\mathscr{F}_n, \mathscr{U}_n, n) \in \mathcal{B}.$$

Then

$$\mathsf{G}_{\text{fin}}(\mathcal{A}',\mathcal{B}') \leqslant_{\text{II}} \mathsf{G}_{\text{fin}}(\mathcal{A},\mathcal{B}).$$

If $\mathsf{T}_{\text{I}}(\cdot, n) = \mathsf{T}_{\text{I}}(\cdot, 1)$ for all $n \in \omega$, then we can strengthen the conclusion to

$$\mathsf{G}_{\text{fin}}(\mathcal{A}',\mathcal{B}') \leqslant^{+}_{\text{II}} \mathsf{G}_{\text{fin}}(\mathcal{A},\mathcal{B}).$$

The analogous result for single-selection games also obtains.

Applications

$\mathsf{G}_1(\Omega_X, \Omega_X) \leqslant_{\mathrm{II}} \mathsf{G}_1(\Omega_{C_p(X),\mathbf{0}}, \Omega_{C_p(X),\mathbf{0}})$

For $\mathscr{F} \in \Omega_{C_p(X),\mathbf{0}}$,

$$\overleftarrow{\mathsf{T}}_{\mathrm{I}}(\mathscr{F}, n) = \{f^{\leftarrow}[(-2^{-n}, 2^{-n})] : f \in \mathscr{F}\}.$$

For $U \in \overleftarrow{\mathsf{T}}_{\mathrm{I}}(\mathscr{F}, n)$, let $\overrightarrow{\mathsf{T}}_{\mathrm{II}}(U, \mathscr{F}, n) \in \mathscr{F}$ be such that

$$U = \overrightarrow{\mathsf{T}}_{\mathrm{II}}(U, \mathscr{F}, n)^{\leftarrow}[(-2^{-n}, 2^{-n})].$$

- Keeping track of the turn number n in $\overleftarrow{\mathsf{T}}_{\mathrm{I}}$ is necessary to guarantee clustering to $\mathbf{0}$.
- Keeping track of $\mathscr{F}$ in $\overrightarrow{\mathsf{T}}_{\mathrm{II}}$ is necessary to guarantee that Two is making legal choices in $\mathsf{G}_1(\Omega_{C_p(X),\mathbf{0}}, \Omega_{C_p(X),\mathbf{0}})$.

We've been able to successfully apply these translation techniques to a variety of contexts:

- The equivalence of $\mathsf{G}_1(\Omega_X, \Omega_X)$ and $\mathsf{G}_1(\Omega_{C_p(X),\mathbf{0}}, \Omega_{C_p(X),\mathbf{0}})$ (and, by using duality, equivalences with other related games like the closed discrete selection game on $C_p(X)$).
- The analogous equivalence of $\mathsf{G}_1(\mathcal{K}_X, \mathcal{K}_X)$ and $\mathsf{G}_1(\Omega_{C_k(X),\mathbf{0}}, \Omega_{C_k(X),\mathbf{0}})$ (and related games, like above).
- Similar single-selection game correspondences involving the classes of minimal usco (upper semicontinuous compact-valued) and minimal cusco (convex-valued usco) maps.
- Equivalence of games in various hyperspace constructions involving cover types defined by Z. Li.
- Some traditional star-selection principles (after noting that star-selection principles can be witnessed as standard selection principles).

Limitations

A commonly used tool in selection principles theory is a bijection $\omega^2 \to \omega$. One would hope to be able to weave in a bijection into the translation schema, but an obstacle presents itself at the level of perfect-information strategies for One.

Let

$$X^{<\omega} = \bigsqcup_{n\in\omega} X^{n+1}$$

and

$$X^{\leqslant n+1} = \bigsqcup_{j=1}^{n+1} X^j$$

for $n \in \omega$.

$\mathsf{G}_\square(\Omega_X, \Omega_X) \leqslant^+_{\mathrm{II}} \mathsf{G}_\square(\Omega_{X^{<\omega}}, \Omega_{X^{<\omega}})$

Fix a bijection $\beta : \omega^2 \to \omega$ which has the property that $\beta(0,0) = 0$ and $\langle \beta(n,k) : k \in \omega \rangle$ is increasing for each $n \in \omega$.

An adaptation of the Wallace Theorem asserts that, for an ω-cover $\mathscr{U}$ of $X^{\leqslant n+1}$, there exists $\mathscr{V}_n(\mathscr{U}) \in \Omega_X$ such that,

$$\forall V \in \mathscr{V}_n(\mathscr{U})\, \exists U \in \mathscr{U} \ \left[V^{\leqslant n+1} \subseteq U\right].$$

So we can define

$$\overleftarrow{\mathsf{T}}_{\mathrm{I}}(\mathscr{U}, \beta(n,k)) = \mathscr{V}_n(\mathscr{U}).$$

We also define

$$\overrightarrow{\mathsf{T}}_{\mathrm{II}}(\mathscr{F}, \mathscr{U}, \beta(n,k)) \in [\mathscr{U}]^{<\omega}$$

such that, for each $V \in \mathscr{F}$, there is $U \in \overrightarrow{\mathsf{T}}_{\mathrm{II}}(\mathscr{F}, \mathscr{U}, \beta(n,k))$ such that $V^{\leqslant n+1} \subseteq U$.

Markov strat. for Two

If Two has a Markov strategy $\overleftarrow{\tau}$ in $\mathsf{G}_{\square}(\Omega_X, \Omega_X)$, then we could define

$$\tau(\mathscr{U}, \beta(n,k)) = \overrightarrow{\mathsf{T}}_{\mathrm{II}}\left(\overleftarrow{\tau}\left(\overleftarrow{\mathsf{T}}_{\mathrm{I}}(\mathscr{U}, \beta(n,k)), k\right), \mathscr{U}, \beta(n,k)\right).$$

Then, for fixed $n \in \omega$, $\langle \overleftarrow{\tau}(\cdot, k)\rangle_{k\in\omega}$ produces an ω-cover of X, which then pushes forward to an ω-cover of $X^{\leqslant n+1}$. Hence, τ is a winning Markov strategy for Two in $\mathsf{G}_{\square}(\Omega_{X^{<\omega}}, \Omega_{X^{<\omega}})$.

Perfect-info. strat. for Two

If Two has a perfect-information strategy in $\mathsf{G}_\square(\Omega_X, \Omega_X)$, a similar line of reasoning shows that they also have a perfect-information strategy in $\mathsf{G}_\square(\Omega_{X^{<\omega}}, \Omega_{X^{<\omega}})$.

(We would apply the strategy on each $\langle \beta(n,k) : k \in \omega \rangle$ and they coalesce into a winning strategy for Two.)

Perfect-info. strat. for One (!)

If we now assume that One doesn't have a winning strategy $\mathsf{G}_\square(\Omega_X, \Omega_X)$, we are assuming that, for any perfect-information strategy σ One tries to employ in that game, there exists a run of the game where σ loses. So we fix a strategy σ for One in $\mathsf{G}_\square(\Omega_X, \Omega_X)$. The idea, as above, is to somehow play the game on the sequences $\langle \beta(n,k) : k \in \omega \rangle$, but this is where we run into a problem...

The Obstacle

Consider $\beta(1,0) > 0$ and suppose we've been concurrently playing the game on $X^{<\omega}$ with two families $\{\mathscr{F}_j : j < \beta(1,0)\}$ and $\{\mathscr{G}_j : j < \beta(1,0)\}$ according to σ. Let

$$\mathscr{U}_j = \sigma\langle \mathscr{F}_\ell : \ell < j\rangle \quad \text{and} \quad \mathscr{V}_j = \sigma\langle \mathscr{G}_\ell : \ell < j\rangle$$

for each $j \leqslant \beta(1,0)$.

It could be the case that $\mathscr{U}_{\beta(1,0)} \neq \mathscr{V}_{\beta(1,0)}$. So which one (of course, there are potentially infinitely many other options) do we choose to be the starting play by One on this branch of $\mathsf{G}_\square(\Omega_X, \Omega_X)$?

Deus Ex Machina

What saves us in this context is Scheepers' theorem that

$$\mathrm{I} \uparrow \mathsf{G}_{\square}(\Omega_X, \Omega_X) \iff \mathrm{I} \underset{\text{pre}}{\uparrow} \mathsf{G}_{\square}(\Omega_X, \Omega_X).$$

Pre-determined strat. for One

One would start with a sequence $\langle \mathscr{U}_n : n \in \omega \rangle$ and Two can make the right kinds of covers using the sequences $\langle U_{\beta(n,k)} : k \in \omega \rangle$.

Constant strat. for One

Essentially identical to the pre-determined case.

Future Investigation

- Is there a general way, other than relying on Hurewicz/Pawlikowski/Scheepers style results, to remedy the translation obstacle when using a bijection $\omega^2 \to \omega$?
- Can these translation techniques apply directly to Hurewicz-style games, or are there "simple" modifications to get them to work there?
- What, if anything, can be said and/or gained from understanding the structure of $\leqslant_{\mathrm{II}}$?

Thank You!